Jimmy Dong

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EDUCATION

B.S. Computer Engineering Technology & Electrical Engineering Technology May '28

Rochester Institute of Technology | Rochester, NY

Coursework: Advanced Electronics, Signals, Systems & Transformers , Hardware Description Language, Electronic Devices, Microcontroller Systems, Digital Systems Design, Circuits I & II.

SKILLS

Hardware: MSP430/MSP432, DE0-CV FPGA, through-hole soldering, surface mount soldering, prototyping, circuit analysis

Programming/HDL: C/C++, VHDL, System Verilog, Verilog, Python, MATLAB, Java, R

Tools: Altium Designer, Quartus, ModelSim, Multisim,

PROJECTS

DEV1 Wire Harness Team Sep '25 - Present

RIT Electric Vehicle Team | Rochester NY

- Integrated and routed wire harness systems to reduce signal interference and improve electrical reliability
- Performed continuity, voltage, and signal validation to verify harness integrity and electrical reliability.
- Assisted in diagnosing and resolving electrical faults, including connector failures and signal inconsistencies within drivetrain and control modules

Automated Texas Hold'em Card-Dispensing Robot (MSP432) Apr '26 - May '26

- Developed an MSP432-based card-dispensing system for Texas Hold'em using Embedded C and a state-machine structure to organize player, discard, flop, turn, and river operations.
- Integrated a TB6612FNG motor driver and DC feeder motor, configuring GPIO and TimerA PWM to control motor direction, speed, and card-feeding duration.
- Programmed and tested low-level GPIO, timer, PWM, and motor-control functions while troubleshooting hardware and firmware integration on the MSP432 platform.

4-Bit Counter Analog-to-Digital Converter (Multisim) Apr '25 - May '25

- Analyzed and troubleshot a 4-bit counter-based A/D converter integrating a 74161 binary counter, op-amp DAC, and LM311 voltage comparator
- Diagnosed and corrected DAC circuit faults to achieve approximately 0.75 V output steps and an 11.25 V full-scale output
- Simulated and verified A/D conversion at a 1 kHz clock frequency across multiple analog input voltages, validating binary output and conversion behavior

Parallel Resonance Band-Pass Filter Design Sep '25 - Oct '25

- Designed and analyzed a parallel resonant band-pass filter using ideal and non-ideal inductor models to determine resonant frequency, Q factor, bandwidth, and voltage gain
- Performed AC frequency sweep simulation to identify resonance and 3 dB cutoff frequencies, comparing theoretical and simulated results
- Built and tested the physical circuit, collecting 50+ frequency response measurements to evaluate the impact of inductor series resistance on Q factor and bandwidth

7-Segment Display Controller (MSP430 Microcontroller) Feb '25 - Apr '25

- Programmed an MSP430 microcontroller in C to drive a multiplexed 4-digit seven-segment display via GPIO and timer-based refresh logic
- Executed digit extraction, timing control, and display refreshing algorithms for real-time numeric output without visible flicker
- Integrated switch inputs and validated display timing across multiple operating conditions